Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/27/2012
PI Name:	Remiker, Ross		
Project Title:	Non-Thermal Sanitation By Atmospheric Pressure Plasma		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	(608) 229-2746
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City:	Madison	**State:**	WI
Zip Code:	53717-1961	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	01/31/2012	**End Date:**	10/23/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed per HRP Technology Pipeline spreadsheet sent by B. Corbin (Ed., 9/9/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX12CA62C		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	ORBITEC's Non-Thermal Sanitation by Atmospheric Pressure Plasma technology sanitizes fresh fruits and vegetables without the use of consumable chemicals and without significantly raising the temperature of the food, so food taste and quality are not affected. Atmospheric pressure plasma is well known to be highly effective in promoting oxidation, enhancing molecular dissociation, and producing free radicals and other types of high energies. It has recently attracted much attention in the food industry due to its potential for being a non-thermal and highly effective sanitation method. The proposed technology will support surface sanitation of delivered fresh fruit and vegetables, and freshly prepared foods in a space-based habitat. It can function in reduced gravity and pressure environments, and is efficient in terms of waste and resource use. During this Phase 2 effort, designs of the primary operating components of the system will be refined and incorporated into a plasma processing chamber prototype capable of treating one to two servings of fresh food at a time. The antimicrobial performance of the prototype will be tested with a number of fruits/vegetables and		

	different inoculums. The prototype will also be evaluated for the effect of plasma treatment on food quality. POTENTIAL NASA COMMERCIAL APPLICATIONS: Non-thermal plasma can be used to sanitize fresh foods grown in the space habitat and to sanitize raw ingredients either produced on orbit or sent up as bulk raw ingredients. Non-thermal plasma does not raise the temperature of the treated food significantly, so it has minimal effect on food quality. Non-thermal plasma can replace chemical disinfectants in most applications. The technology has low system mass, consumable mass, and power consumption. The function of this technology is insensitive to reduced cabin pressure and reduced gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Terrestrially, non-thermal plasma sanitation technology can reduce the number of cases of food-borne illness due to contaminations by sanitizing food at the point of use, such as a restaurant, or at a food processing facility. This technology can be used in place of chlorinated water, which can leave a residue and is not entirely effective, and irradiation, which generally has a poor public perception.
Task Progress:	New project for FY2012. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)